

Telecom Customer Churn Analysis

Developed by Nakshathira S
Power BI | DAX | Data Analytics

1869

CUSTOMERS AT RISK

2173

OF TECH TICKETS

885

ADMIN TICKETS

\$2.86M

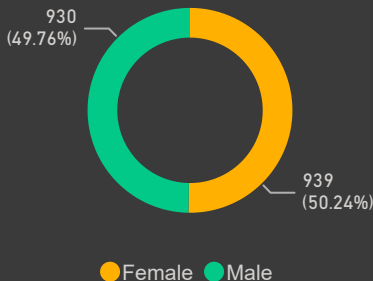
YEARLY CHARGES

139.13K

MONTHLY CHARGES

DEMOGRAPHICS BY GENDER

CHURNED CUSTOMERS BY GENDER



25%

Senior Citizen

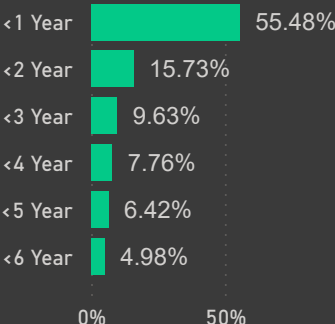
36%

Partner

17%

Dependents

SUBSCRIPTION TIME



17%

Tech Support

44%

Streaming TV

44%

Streaming Movies

29%

Device Protection

28%

Online Backup

16%

Online Security

91%

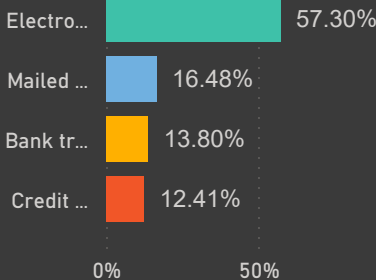
Phone Service

Multiple Lines

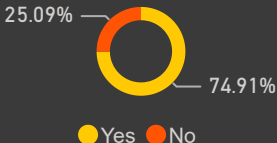
50.03%
Yes

49.97%
No

PAYMENT METHOD



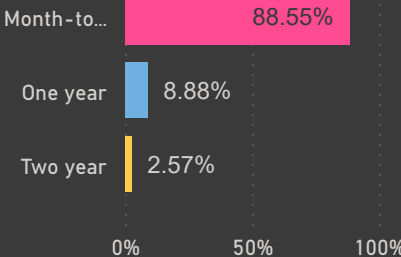
PAPERLESS BILLING



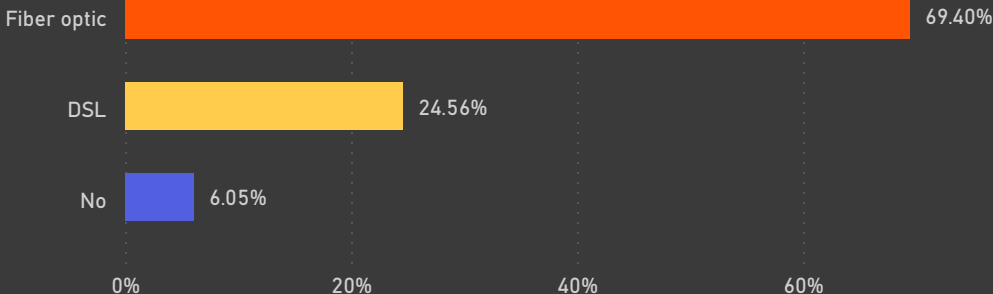
AVG CHARGES

Monthly
Total

TYPE OF CONTRACTS



INTERNET SERVICE USERS



Customer Retention & Risk Analysis

CHURN

All

INTERNET SERVICE

All

TENURE

0 72

CONTRACT

All

7043

Total Customers

26.54%

Churn Rate

\$16.06M

Yearly Charges

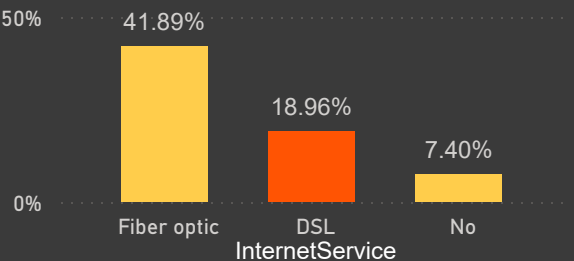
3632

Admin Tickets

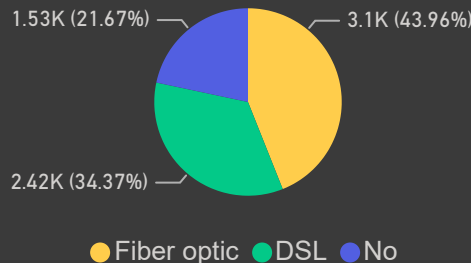
2955

Tech Tickets

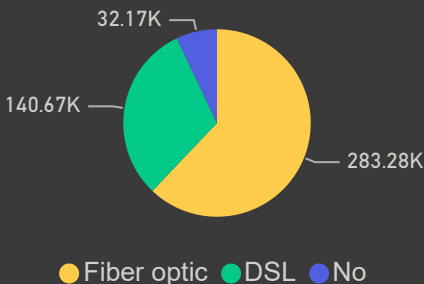
Churn Rate BY Type of Internet Service



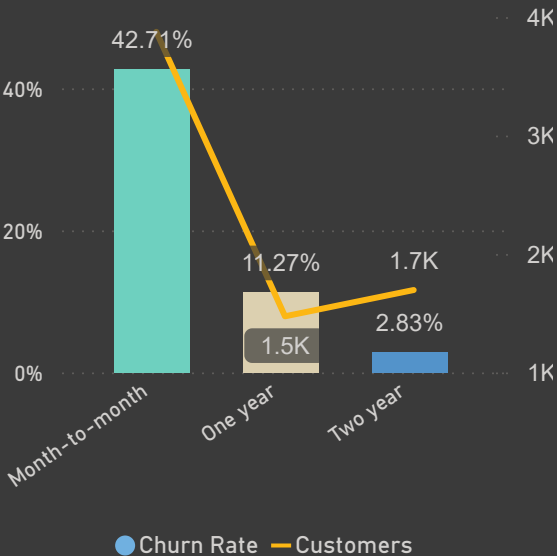
Of Customers by Internet Service



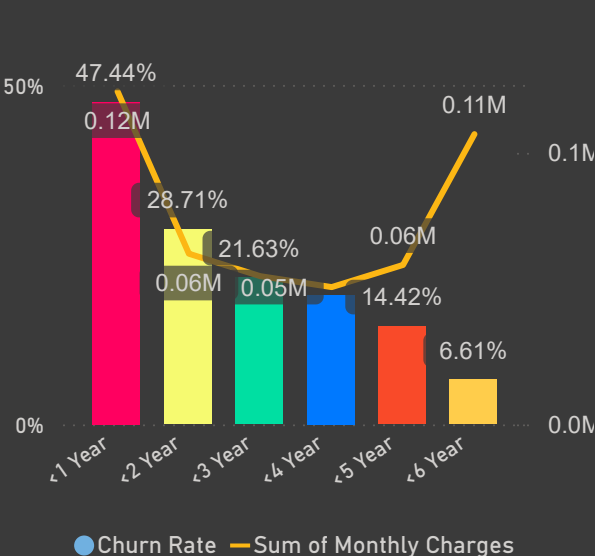
Sum of Monthly Charges



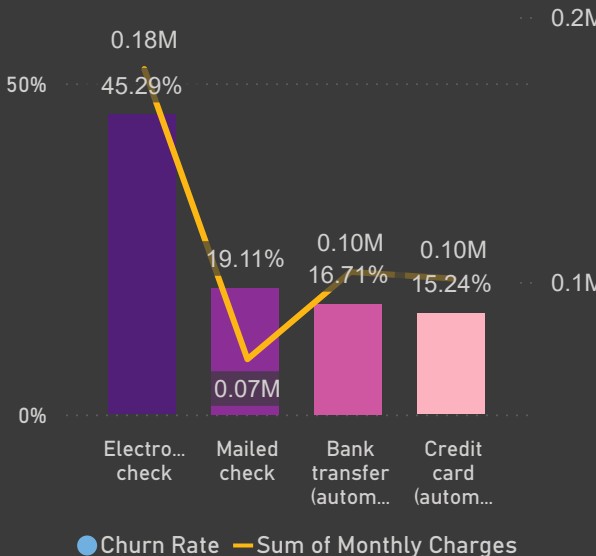
Type Of Contract



Years of Contract



Churn by Payment Method



KEY INSIGHTS

- The customer churn rate last month was 27%, which means that out of 7043 customers, 1869 left the company.
- The length of the contract and the tenure of the customer are key factors in predicting the churn behavior. Customers who have a monthly contract and a lower tenure are more likely to switch to other providers.
- Customers who do not have any dependents or partners are more prone to churn than those who do.
- Gender does not seem to have a significant impact on the churn decision. However, senior citizens are less likely to churn than non-senior citizens.
- Customers who use Fibre Optic internet service have a higher churn rate than those who use other types of internet service. The payment method also influences the churn decision, with Electronic check being the most common among the churned customers.
- Customers who do not receive services such as Tech Support, Device Protection, and Online Security are more dissatisfied and tend to look for other options.

Email

Dear Bob,

I have completed the customer churn analysis and developed an interactive Power BI dashboard to identify the major factors influencing customer attrition. The analysis provides valuable insights into customer behavior and highlights opportunities to improve retention.

Key Findings:

- The overall customer churn rate is 26.54%, with 1,869 customers leaving out of a total of 7,043 customers.
- Customers with month-to-month contracts exhibit the highest churn rate compared to one-year and two-year contract customers.
- Customers with shorter tenure are more likely to leave the company, while long-term customers demonstrate greater loyalty.
- Fiber Optic internet users show a significantly higher churn rate than DSL and non-internet customers.
- Electronic Check is the most common payment method among customers who churn.
- Customers without services such as Tech Support, Online Security, and Device Protection are more likely to discontinue their subscriptions.
- Gender has minimal impact on churn behavior, whereas customer tenure and contract type are strong indicators of retention.

Recommendations:

- Encourage customers to move from month-to-month contracts to longer-term plans through discounts and loyalty benefits.
- Introduce retention campaigns for new customers during their initial months of service.
- Bundle value-added services such as Tech Support, Online Security, and Device Protection into standard subscription plans.
- Develop targeted offers for Fiber Optic customers to improve satisfaction and reduce churn risk.
- Create personalized loyalty programs for high-risk customer segments identified through the dashboard.

The dashboard enables stakeholders to monitor customer retention trends, identify churn risk factors, and make data-driven business decisions to improve customer satisfaction and long-term revenue.

Thank you for your time and consideration.

Best Regards,
Nakshathira S
Power BI Developer & Data Analyst